

Paper 1 - Set 15 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Trigonometric functions and equations
2	D	Number, units and estimation
3	F	GCSE algebra
4	D	Polynomials, factor and remainder theorems
5	C	Triangles, radians and sectors
6	B	Exponentials and logarithms
7	B	Circles and circle theorems
8	E	Graphs of functions
9	B	Pythagoras, mensuration and 3D
10	C	Triangles, radians and sectors
11	B	Pythagoras, mensuration and 3D
12	B	Quadratic functions
13	B	Straight lines
14	E	Indices and surds
15	F	Straight lines
16	B	Indices and surds
17	E	Differentiation
18	C	Sequences and series
19	A	Binomial expansion
20	B	Polynomials, factor and remainder theorems

Worked solutions

Question 1 TRG-06 · Trigonometric functions and equations · spec MM4.5

Given that $\sin \theta = \frac{3}{5}$ and θ is obtuse, find $\tan \theta$.

Answer: A

$$\cos \theta = -\frac{4}{5} \text{ (negative in the second quadrant), so } \tan \theta = \frac{3/5}{-4/5} = -\frac{3}{4}.$$

Question 2 NUM-04 · Number, units and estimation · spec M2.13

A length is given as 4.6 cm correct to 1 decimal place, and a width as 3 cm correct to the nearest cm. Find the upper bound of the area.

Answer: D

$$\text{Upper bounds 4.65 and 3.5: } 4.65 \times 3.5 = 16.275.$$

Question 3 ALG-11 · GCSE algebra · spec M4.4

Expand and simplify $(x + 2)(x - 1)(x + 3)$.

Answer: F

$$(x^2 + x - 2)(x + 3) = x^3 + 3x^2 + x^2 + 3x - 2x - 6 = x^3 + 4x^2 + x - 6.$$

Question 4 POL-15 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

Answer: D

$$p(1) = 1 + a + b + c, \text{ and } p(1) = (1 - 2)(1 + 1)(1 - 3) = (-1)(2)(-2) = 4. \text{ So } a + b + c = 3. \text{ (Directly: } (x - 2)(x + 1)(x - 3) = x^3 - 4x^2 + x + 6.)$$

Question 5 TRI-03 · Triangles, radians and sectors · spec MM4.2

A sector of a circle of radius 6 cm has area $15\pi \text{ cm}^2$. Find the arc length of the sector.

Answer: C

$$\frac{1}{2}r^2\theta = 18\theta = 15\pi \Rightarrow \theta = \frac{5\pi}{6}. \text{ Arc} = r\theta = 6 \cdot \frac{5\pi}{6} = 5\pi.$$

Question 6 LOG-02 · Exponentials and logarithms · spec MM5.3

Solve $3^{2x} = 4$. Give x in terms of logarithms to base 3.

Answer: B

$$2x = \log_3 4 = 2\log_3 2, \text{ so } x = \log_3 2.$$

Question 7 CIR-03 · Circles and circle theorems · spec MM3.2

The circle with centre $(2, -1)$ passes through $(5, 3)$. Find the equation of the tangent to the circle at $(5, 3)$.

Answer: B

$$\text{Radius direction } (3, 4) \text{ has gradient } \frac{4}{3}, \text{ so the tangent has gradient } -\frac{3}{4}; \\ y - 3 = -\frac{3}{4}(x - 5) \Rightarrow 4y - 12 = -3x + 15 \Rightarrow 3x + 4y = 27.$$

Question 8 GRA-12 · Graphs of functions · spec MM8.1

Which of the following functions is undefined at exactly two real values of x ?

Answer: E

$x^2 - 4 = 0$ at $x = \pm 2$: two values. $x^2 + 4$ is never zero; $\frac{x}{x^2}$ and $\frac{1}{(x-2)^2}$ are undefined at one value; $\sqrt{x-2}$ is undefined for all $x < 2$.

Question 9 MEN-01 · Pythagoras, mensuration and 3D · spec M5.7

A cuboid measures 3 cm by 4 cm by 12 cm. Find the length of its longest diagonal.

Answer: B

$$\sqrt{9 + 16 + 144} = \sqrt{169} = 13.$$

Question 10 TRI-05 · Triangles, radians and sectors · spec MM4.2

Convert 135° to radians.

Answer: C

$$135 = \frac{3}{4} \times 180, \text{ so } \frac{3\pi}{4}.$$

Question 11 MEN-12 · Pythagoras, mensuration and 3D · spec M5.16

A running track consists of two straights of length 100 m and two semicircular ends of radius r . The total length of the track is 400 m. Find r .

Answer: B

$$200 + 2\pi r = 400 \Rightarrow r = \frac{100}{\pi}.$$

Question 12 QUA-07 · Quadratic functions · spec MM1.3

How many real solutions does the equation $x^4 - 5x^2 + 4 = 0$ have?

Answer: B

Let $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4) = 0$, so $x^2 = 1$ or $x^2 = 4$, giving $x = \pm 1, \pm 2$: four solutions.

Question 13 LIN-13 · Straight lines · spec MM3.1

The distance between the parallel lines $3x + 4y = 5$ and $3x + 4y = 20$ is

Answer: B

The lines are $3x + 4y = c$; the perpendicular direction is $(3, 4)$ of length 5. Distance $= \frac{|20-5|}{5} = 3$. (Alternatively, the point $(0, \frac{5}{4})$ on the first line is 3 from the second.)

Question 14 IND-06 · Indices and surds · spec MM1.2

Given that $(2 + \sqrt{3})^2 = a + b\sqrt{3}$ and $(2 + \sqrt{3})^{-1} = c + d\sqrt{3}$, where a, b, c, d are integers, find the value of $a + b + c + d$.

Answer: E

$$(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}, \text{ so } a = 7, b = 4. (2 + \sqrt{3})^{-1} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}, \text{ so } c = 2, d = -1. \text{ Sum} \\ = 7 + 4 + 2 - 1 = 12.$$

Question 15 LIN-11 · Straight lines · spec MM3.1

The vertices of a triangle are $A(0, 0)$, $B(6, 0)$ and $C(2, 4)$. Find the equation of the line through C that bisects the area of the triangle.

Answer: F

A line through a vertex bisects the area iff it passes through the midpoint of the opposite side, $(3, 0)$. Through $(2, 4)$ and $(3, 0)$: gradient -4 , $y = -4x + 12$.

Question 16 IND-11 · Indices and surds · spec MM1.2

Given that $x = \sqrt{7} + \sqrt{3}$ and $y = \sqrt{7} - \sqrt{3}$, find $x^2 + y^2 - xy$.

Answer: B

$x^2 + y^2 = (10 + 2\sqrt{21}) + (10 - 2\sqrt{21}) = 20$ and $xy = 7 - 3 = 4$. So $20 - 4 = 16$.

Question 17 DIF-15 · Differentiation · spec MM6.3

The volume of a cuboid with a square base of side x is $V = 4x^2 - x^3$ for $0 < x < 4$. Find the maximum volume.

Answer: E

$V' = 8x - 3x^2 = x(8 - 3x) = 0 \Rightarrow x = \frac{8}{3}$. $V = \frac{64}{9} \cdot 4 - \frac{512}{27} = \frac{768 - 512}{27} = \frac{256}{27}$.

Question 18 SEQ-17 · Sequences and series · spec MM2.3

A ball is dropped from a height of 10 m. After each bounce it rises to $\frac{3}{5}$ of its previous height. Find the total distance travelled by the ball before it comes to rest.

Answer: C

Down: $10 + 6 + 3.6 + \dots = \frac{10}{1 - 3/5} = 25$. Up: $6 + 3.6 + \dots = 15$. Total 40 m.

Question 19 BIN-11 · Binomial expansion · spec MM2.4

Find the value of $(\sqrt{2} + 1)^4 + (\sqrt{2} - 1)^4$.

Answer: A

Odd powers of $\sqrt{2}$ cancel: $2[(\sqrt{2})^4 + \binom{4}{2}(\sqrt{2})^2 + 1] = 2[4 + 12 + 1] = 34$.

Question 20 POL-06 · Polynomials, factor and remainder theorems · spec MM1.6

When $p(x)$ is divided by $(x - 1)$ the remainder is 3, and when divided by $(x + 1)$ the remainder is -1 . Find the remainder when $p(x)$ is divided by $(x^2 - 1)$.

Answer: B

The remainder has degree < 2 : $ax + b$ with $a + b = 3$ and $-a + b = -1$. So $b = 1$, $a = 2$: remainder $2x + 1$.